

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)		Acceleration (gradient) or equivalent = $\frac{(v-u)}{t}$ (1) Displacement (area) = $ut + \frac{1}{2}t(v-u)$ or $\frac{(u+v)}{2}t$ (1) Substitution and clear algebra seen e.g. $x = ut + \frac{1}{2}t(at)$ seen (1) No reference to graph award a maximum of 2 marks	3			3	3	
	(b)	(i)	11 sin50° [= 8.4 [3] m s ⁻¹]		1		1	1	
		(ii)	Attempt at substituting into $H = ut + \frac{1}{2}at^2$ even if incorrect signs (1) $H = 8.4 \times 1.95 - \frac{1}{2} \times 9.81 \times (1.95)^2$ seen (1) Award 2 marks if only this seen $H = -2.27$ [m] (1) accept 2.21 [m] Accept use of $v = 8$ m s⁻¹ leading to $H = 3.05$ [m]. Alternative solution: Time to reach max height = $\frac{(0-8.4)}{-9.81}$ (or 0.86 s) (1) Time from H to floor = $1.95 - (2 \times 0.86)$ (or 0.232 s) (1) $H = (8.4 \times 0.23) + \frac{1}{2} \times 9.81 \times (0.23)^2$ (or 2.21 m) (1) Consider carefully consequence of 'rounding'	1 1	1		3	3	
		(iii)	Horizontal component of velocity = $11\cos 50^\circ$ [= 7.1 m s⁻¹] or by Pythagoras (1) Substitution into $v_H = \frac{R}{t}$ $R = 7.1$ (ecf) $\times 1.95 = 13.8$ [m] (1)		2			2	